

Title

ZPL_Macro_Apply "Make Thermal" calculations to Monte Carlo files

Author

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Brief Description

This macro will cycle through Monte Carlo files and apply the Make Thermal tool to them.

Overview

This macro allows users to apply the "Make Thermal" tool to already-generated Monte Carlo files. The amount of MC files is hard coded into the code. This will apply five Multi-Configuration operands per surface: CRVT, THIC, SDIA, CHZN, MCSD. MC operands for extra data parameters such as those for Even Asphere and Irregular surfaces are not included. (Update V2 – Even Aspheres are now included. Other surfaces with extra data parameters are not)

Supplemental Content

Slides from a previous OpticsTalk are also included. These slides walk through the macro's steps in more detail and offer information about how the macro can be further updated.

Plugin Type

ZPL Extension

Language

ZPL

Updates

Date	Version	OpticStudio version	Comment
2019/10/15	1.0	19.4SP2	Creation
2020/08/18	2.0	20.2	• Updates to formatting • Addition of option to choose not to clear the MCE before applying the Make Thermal tool • Addition of a loop which will check for the Even Asphere type. This will add parameters to the MCE which correspond

			to the asphere's extra data parameters.
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Instructions

1. Installation

Copy the macro to the {Zemax}\Macros folder

2. How to use

In OpticStudio, navigate to **Programming...Edit/Run** and open the macro for editing. Change the "path\$", "prefix\$", and "number_of_files" variables to match with the location and number of the Monte Carlo files you want to apply the change to. (Note: this portion of the macro is discussed in the Knowledgebase article "[How to open consecutively-named lens files using a ZPL macro](#)").

```
1! Written 10/15/19
2! This macro will emulate the Make Thermal tool for use in Monte Carlo analysis
3
4! Input temperature data
5INPUT "Enter number of configurations", configNum
6! configNum = 0      # uncomment this line and comment the above for automation
7INPUT "Enter minimum temperature", minTemp
8! minTemp = 0       # uncomment for automation
9INPUT "Enter maximum temperature", maxTemp
10! maxTemp = 0      # uncomment for automation
11
12tempStep = (maxTemp - minTemp) / (configNum - 1)
13
14!!!!!!!!!!!!!! Open file !!!!!!!!!!!!!!!
15path$ = "C:\Documents\"
16prefix$ = "MC_T"
17number_of_files = 20
18
19IF (number_of_files > 9999) THEN number_of_files = 9999
20
21FORMAT "%04i" LIT
22
23FOR index, 1, number_of_files, 1      # NOTE: the "NEXT" for this loop is at the end of the macro
24    suffix$ = $STR(index)
25
26    myfileName$ = path$ + prefix$ + suffix$ + ".zmx"
27    LOADLENS myfileName$
28
```

Save and run the macro. Each Monte Carlo file will then be made into a Multi-Configuration file with thermal data applied:

Code Exchange

Zemax

	Active : 1/4	Config 1'	Config 2	Config 3	Config 4	
1	TEMP ▾ -	20.00...	20.00000000	60.00000000	100.00000000	^
2	PRES ▾ -	1.000...	1.00000000	1.00000000	1.00000000	
3	CRVT ▾ 1	0.000...	0.00000000 T	0.00000000 T	0.00000000 T	
4	THIC ▾ 1	30.00...	30.00000000 T	30.00000000 T	30.00000000 T	
5	SDIA ▾ 1	7.000...	7.00000000 T	7.00000000 T	7.00000000 T	
6	CHZN ▾ 1	0.000...	0.00000000 T	0.00000000 T	0.00000000 T	
7	MCSD ▾ 1	7.000...	7.00000000 T	7.00000000 T	7.00000000 T	
8	CRVT ▾ 2	0.000...	0.00000000 T	0.00000000 T	0.00000000 T	
9	THIC ▾ 2	0.000...	0.00000000 T	0.00000000 T	0.00000000 T	
10	SDIA ▾ 2	0.000...	0.00000000 T	0.00000000 T	0.00000000 T	
11	CHZN ▾ 2	0.000...	0.00000000 T	0.00000000 T	0.00000000 T	
12	MCSD ▾ 2	0.000...	0.00000000 T	0.00000000 T	0.00000000 T	
13	CRVT ▾ 3	0.000...	0.00000000 T	0.00000000 T	0.00000000 T	
14	THIC ▾ 3	0.000...	0.00000000 T	0.00000000 T	0.00000000 T	
15	SDIA ▾ 3	0.000...	0.00000000 T	0.00000000 T	0.00000000 T	
16	CHZN ▾ 3	0.000...	0.00000000 T	0.00000000 T	0.00000000 T	
17	MCSD ▾ 3	0.000...	0.00000000 T	0.00000000 T	0.00000000 T	
18	CRVT ▾ 4	0.028...	0.02879601 T	0.02879601 T	0.02879601 T	
19	THIC ▾ 4	4.146...	4.14617227 T	4.14617227 T	4.14617227 T	
20	SDIA ▾ 4	9.000...	9.00000000 T	9.00000000 T	9.00000000 T	